Worksheet 6

Problem 6.1

```
library("mlbench")
data(Zoo)
#str(Zoo)
dim(Zoo)
```

```
## [1] 101  17
```

```
Xmatrix<-Zoo[,1:16]
Ymatrix<-factor(Zoo[,17])
#str(Xmatrix);str(Ymatrix)

colnames(Xmatrix)
```

```
##  [1] "hair"     "feathers" "eggs"     "milk"     "airborne" "aquatic"
##  [7] "predator" "toothed"  "backbone" "breathes" "venomous" "fins"
## [13] "legs"     "tail"     "domestic" "catsize"
```

```
levels(Ymatrix)
```

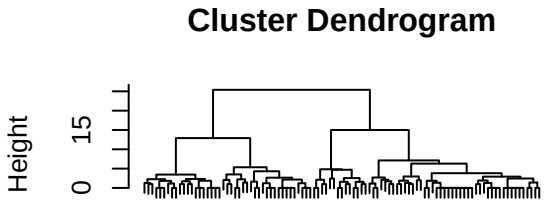
```
## [1] "mammal"        "bird"          "reptile"       "fish"
## [5] "amphibian"     "insect"        "mollusc.et.al"
```

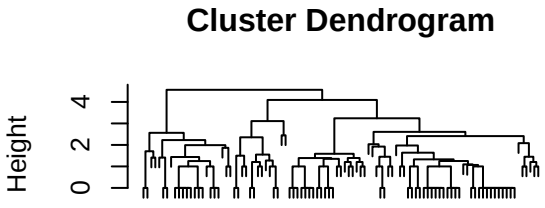
```
summary(Ymatrix)
```

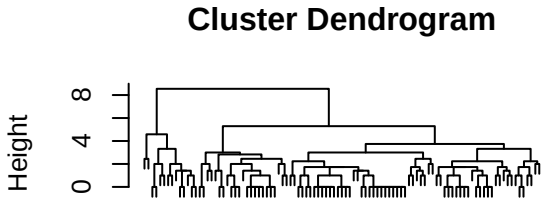
```
##        mammal          bird       reptile          fish     amphibian
##            41            20             5            13             4
##        insect mollusc.et.al
##             8            10
```

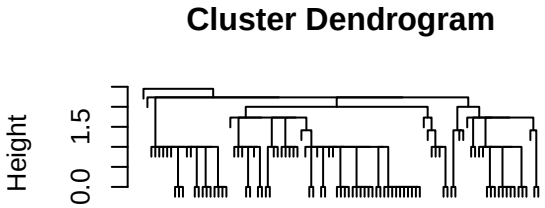
```
distance<-dist(Xmatrix)
hc1<-hclust(distance,method="ward.D2")
hc2<-hclust(distance,method="average")
hc3<-hclust(distance,method="complete")
hc4<-hclust(distance,method="single")
#str(hc1);str(hc2);str(hc3);str(hc4)

par(mfrow=c(2,2))
plot(hc1,label=FALSE)
plot(hc2,label=FALSE)
plot(hc3,label=FALSE)
plot(hc4,label=FALSE)
```

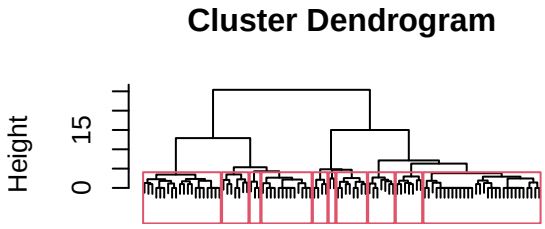


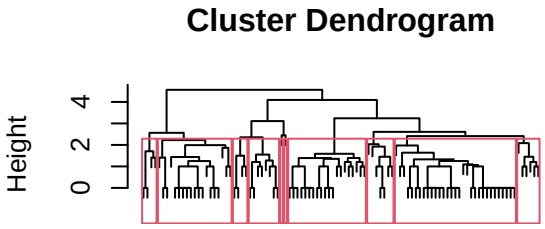


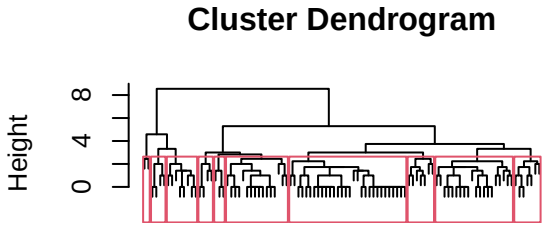


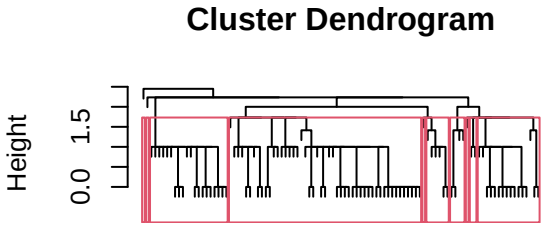



```
tester<-function(h1=NULL,k1=NULL){
    par(mfrow=c(2,2))
    plot(hc1,label=FALSE)
    rect.hclust(tree=hc1,k=k1,h=h1)
    plot(hc2,label=FALSE)
    rect.hclust(tree=hc2,k=k1,h=h1)
    plot(hc3,label=FALSE)
    rect.hclust(tree=hc3,k=k1,h=h1)
    plot(hc4,label=FALSE)
    rect.hclust(tree=hc4,k=k1,h=h1)
    par(mfrow=c(1,1))
}
tester(k1=10)
```



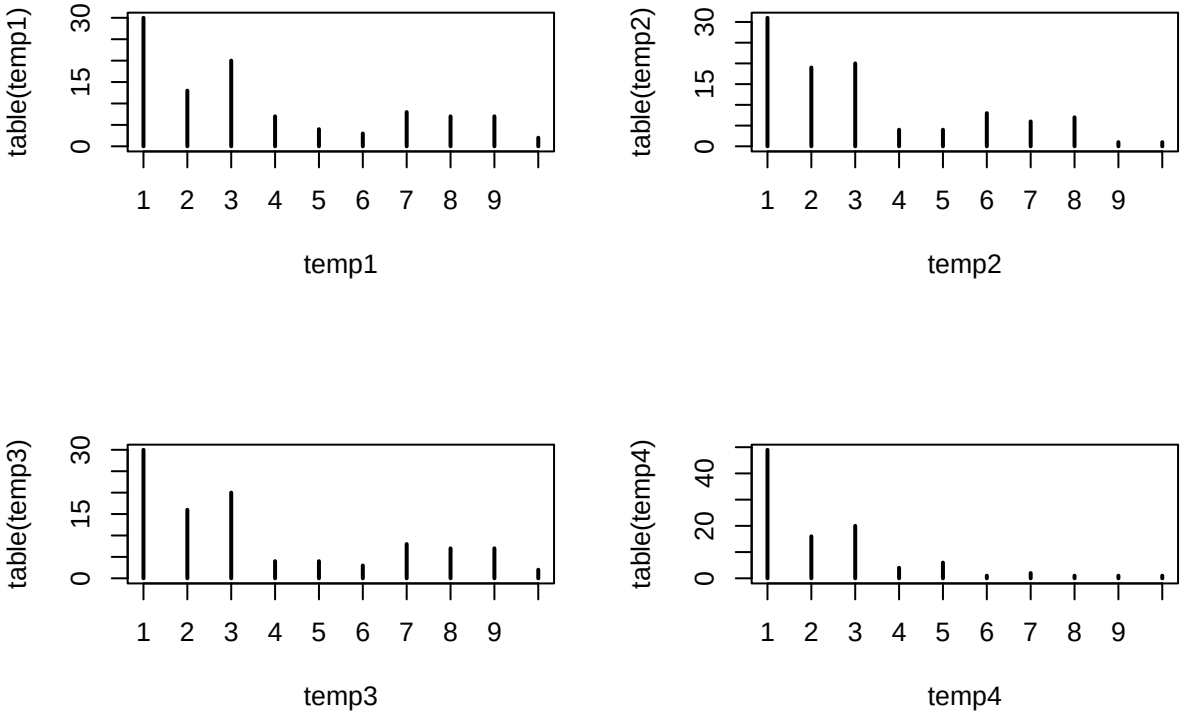







```
tester2<-function(k1=NULL,h1=NULL){
    temp1<-cutree(hc1,k=k1,h=h1)
    temp2<-cutree(hc2,k=k1,h=h1)
    temp3<-cutree(hc3,k=k1,h=h1)
    temp4<-cutree(hc4,k=k1,h=h1)
    print(table(temp1));print(table(temp2))
    print(table(temp3));print(table(temp4))
    par(mfrow=c(2,2))
    plot(table(temp1));plot(table(temp2))
    plot(table(temp3));plot(table(temp4))
}
tester2(k1=10)
```

```
## temp1
##  1  2  3  4  5  6  7  8  9 10
## 30 13 20  7  4  3  8  7  7  2
## temp2
##  1  2  3  4  5  6  7  8  9 10
## 31 19 20  4  4  8  6  7  1  1
## temp3
##  1  2  3  4  5  6  7  8  9 10
## 30 16 20  4  4  3  8  7  7  2
## temp4
##  1  2  3  4  5  6  7  8  9 10
## 49 16 20  4  6  1  2  1  1  1
```



The function "tester" plots the clustering trees and shows which observations belong to which cluster. The number of clusters is determined by the value of k or h: k specifies the number of clusters, while h specifies the height at which the tree is cut. The number of observations in each cluster depends on the tree and where it is cut.

The function "tester2" prints the number of observations in each cluster for each of the four clustering trees.

In my results, the maximum tree height increases from single to average to complete to Ward.D2. This ordering depends on the data and distance measure and may differ for other datasets. Single linkage uses the minimum distance between observations in two clusters, complete linkage uses the maximum distance, and average linkage uses the mean distance. Ward.D2 merges clusters based on the increase in within-cluster sum of squares; it does not use the sum of total distances.

For 10 clusters, the tables show similar cluster sizes across the four methods, with small differences in the number of observations per cluster. This suggests that the cluster-size distributions are similar, but it does not show whether the same observations are in the same clusters. The methods use the same distance matrix but different linkage rules.

Problem 6.2

We want to generate three classes of observations, each assumed to follow a multivariate normal distribution.

```
set.seed(1)
library("mnormt")
n1<-300
mu1<-c(5,5)
sigma1<-matrix(c(1,0,0,1),nrow=2,ncol=2)
```

```r
x1<-rmnorm(n1,mean=mu1,varcov=sigma1)

n2<-500
mu2<-c(8,3)
sigma1<-matrix(c(4,0,0,9),nrow=2,ncol=2)
x2<-rmnorm(n2,mean=mu2,varcov=sigma1)

n3<-900
mu3<-c(12,9)
sigma1<-matrix(c(8,0,0,3),nrow=2,ncol=2)
x3<-rmnorm(n3,mean=mu3,varcov=sigma1)

Xmatrix<-rbind(x1,x2,x3)
Ymatrix<-factor(c(rep("Class A",n1),rep("Class B",n2),rep("Class C",n3)))
head(Ymatrix)
```

```
## [1] Class A Class A Class A Class A Class A Class A
## Levels: Class A Class B Class C
```

```r
temp<-kmeans(Xmatrix,centers=3)
names(temp)
```

```
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```r
temp$centers #mean value of each cluster
```

```
##        [,1]     [,2]
## 1  9.550910 8.914362
## 2  6.625579 3.356202
## 3 14.344397 8.972960
```

```r
temp$size #number of observations in each cluster
```

```
## [1] 488 735 477
```

```r
temp$totss #SSE across all observations
```

```
## [1] 44731.13
```

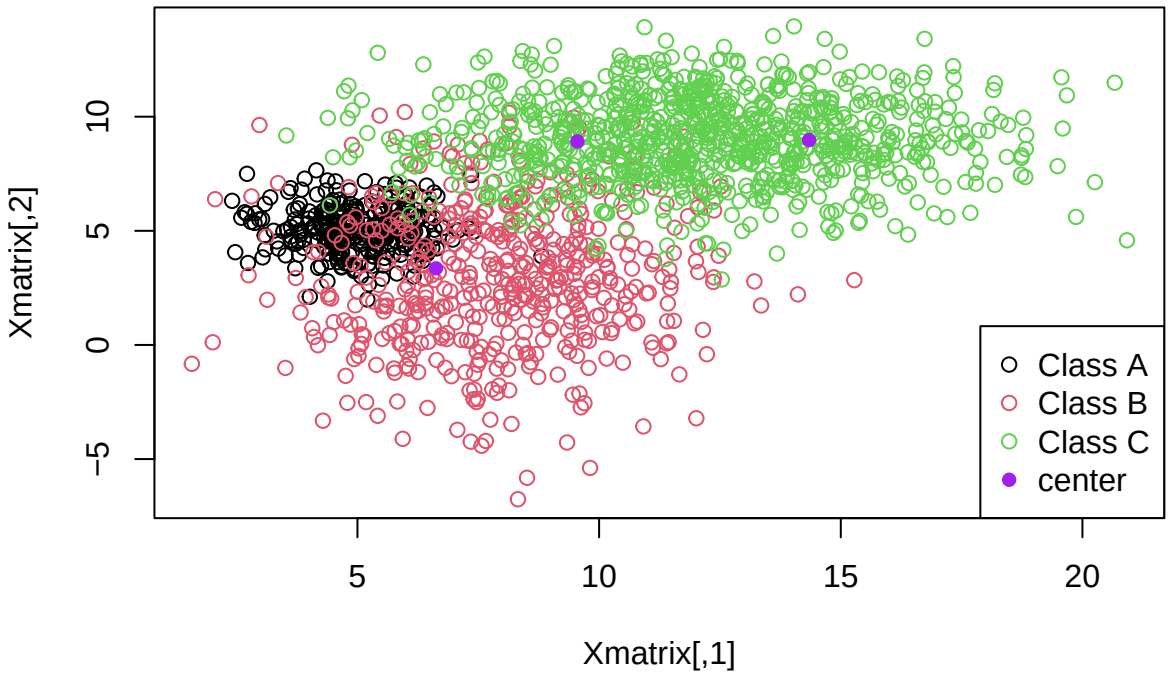
```r
temp$tot.withinss #SSE across all observations in each cluster
```

```
## [1] 14467.21
```

```r
temp$iter #Number of iterations used
```

```
## [1] 2
```

```r
center<-temp$centers
plot(Xmatrix,col=as.integer(Ymatrix),pch=1)
points(center,pch=16,col="purple")
legend("bottomright",legend=c(levels(Ymatrix),"center"),col=c(1,2,3,"purple"),pch=c(1,1,1,16))
```




```
center
```

```
##        [,1]     [,2]
## 1  9.550910 8.914362
## 2  6.625579 3.356202
## 3 14.344397 8.972960
```

```
temp2<-as.matrix(temp$centers)
temp3<-t(temp2)
temp4<-matrix(0,nrow=nrow(temp3),ncol=ncol(temp3))
temp5<-temp4
x<-order(temp3[1,])
for (i in 1:3){
    temp4[2,x[i]]<-temp3[2,i]
    temp4[1,x[i]]<-temp3[1,i]
}
print(temp4)
```

```
##          [,1]     [,2]     [,3]
## [1,] 6.625579 9.550910 14.34440
## [2,] 3.356202 8.914362  8.97296
```

```
temp5<-cbind(mu1,mu2,mu3)
temp5
```

```
##      mu1 mu2 mu3
## [1,]   5   8  12
## [2,]   5   3   9
```

```
temp5-temp4
```

```
##            mu1       mu2        mu3
## [1,] -1.625579 -1.550910 -2.3443974
## [2,]  1.643798 -5.914362  0.0270398
```

```
table(Ymatrix,temp$cluster)
```

```
##
## Ymatrix     1   2   3
##   Class A   1 299   0
##   Class B  69 421  10
##   Class C 418  15 467
```

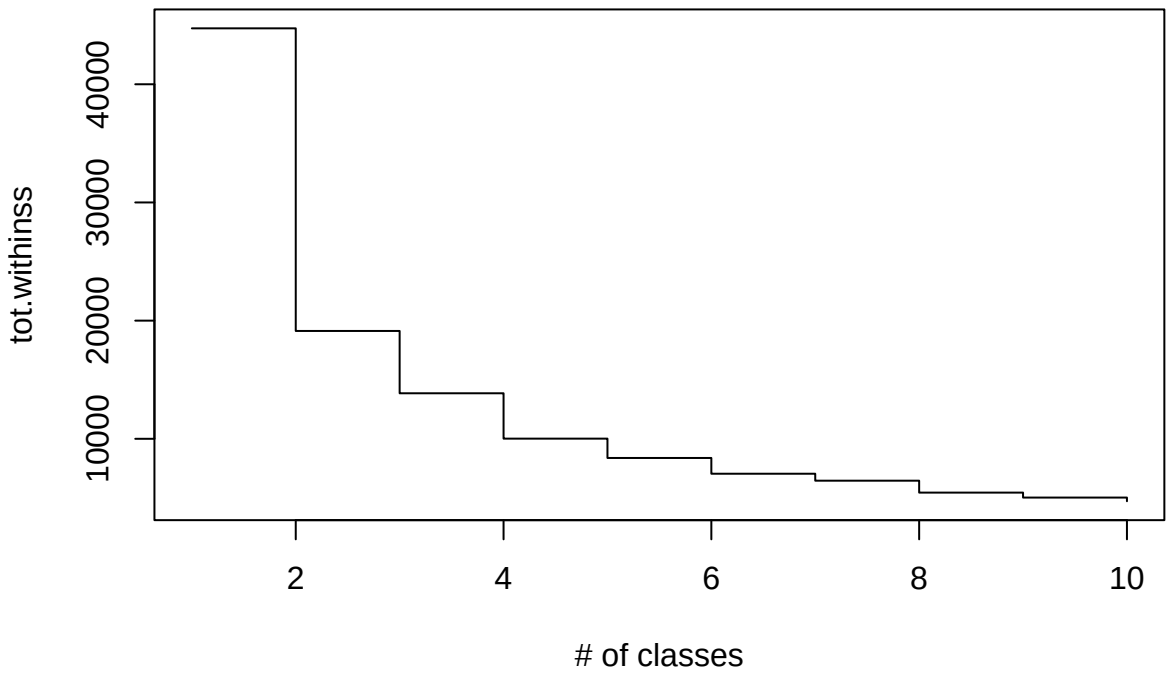
```
names(temp)
```

```
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

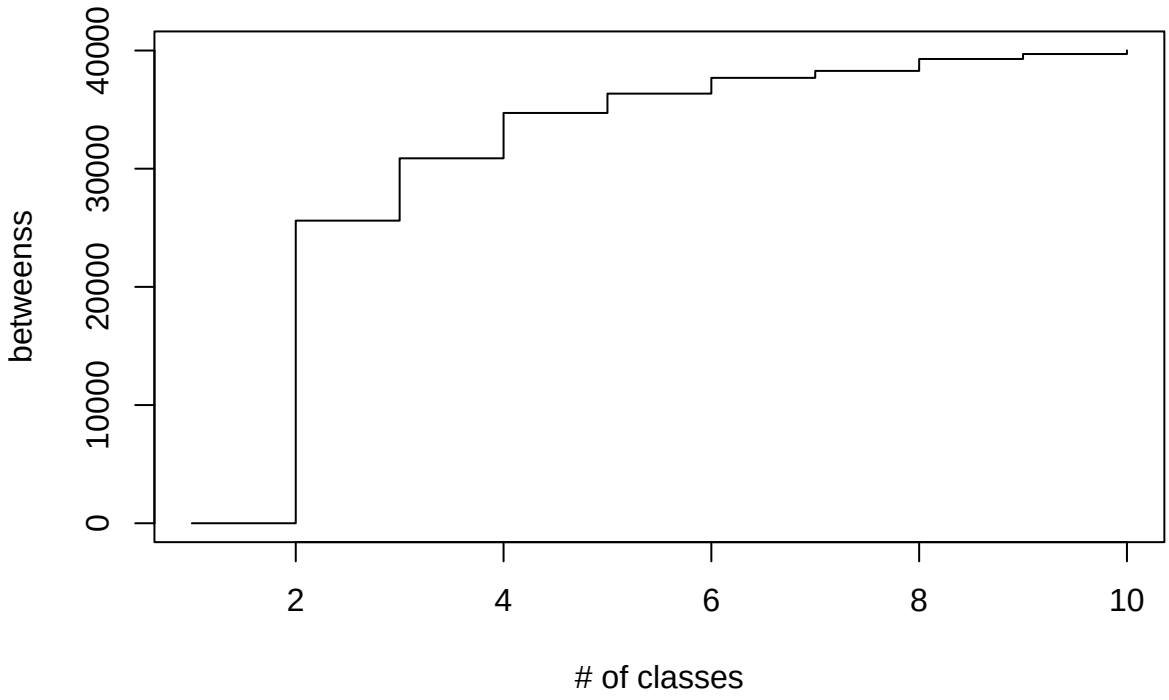
```
temp6<-matrix(0,nrow=10,ncol=2)
for (i in 1:10){
    temp<-kmeans(Xmatrix,centers=i)
    temp6[i,1]<-temp$tot.withinss
    temp6[i,2]<-temp$betweenss
}
colnames(temp6)<-c("tot.withinss","betweenss")
print(temp6)
```

```
##       tot.withinss    betweenss
##  [1,]    44731.132 3.637979e-11
##  [2,]    19124.794 2.560634e+04
##  [3,]    13852.692 3.087844e+04
##  [4,]    10014.344 3.471679e+04
##  [5,]     8378.274 3.635286e+04
##  [6,]     7045.529 3.768560e+04
##  [7,]     6454.888 3.827624e+04
##  [8,]     5446.948 3.928418e+04
##  [9,]     5026.851 3.970428e+04
## [10,]     4716.709 4.001442e+04
```

```
plot(1:10,temp6[,1],xlab="# of classes",ylab="tot.withinss",type="s")
```




```
plot(1:10,temp6[,2],xlab="# of classes",ylab="betweenss",type="s")
```



The center of each cluster is different from the true mean of each class. This may be due to overlap between the classes and their different variances.

The number of observations in each cluster is also different from the true number of observations in each class, especially for Class C. This is because K-means does not make the cluster sizes match the original class sizes.

As the number of clusters increases, the total within-cluster variation decreases, but the between-cluster variation increases. This is because more clusters allow observations to be assigned to closer centers, reducing the variation within each cluster. The total variation stays the same, so the decrease in within-cluster variation corresponds to an increase in between-cluster variation.

Problem 6.3

We want to apply the same analytical methods as in Problem 6.2, but to real-world data rather than generated data.

```
set.seed(1)
data(iris)
str(iris)
```

```
## 'data.frame':    150 obs. of  5 variables:
##  $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
##  $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
##  $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
##  $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
##  $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

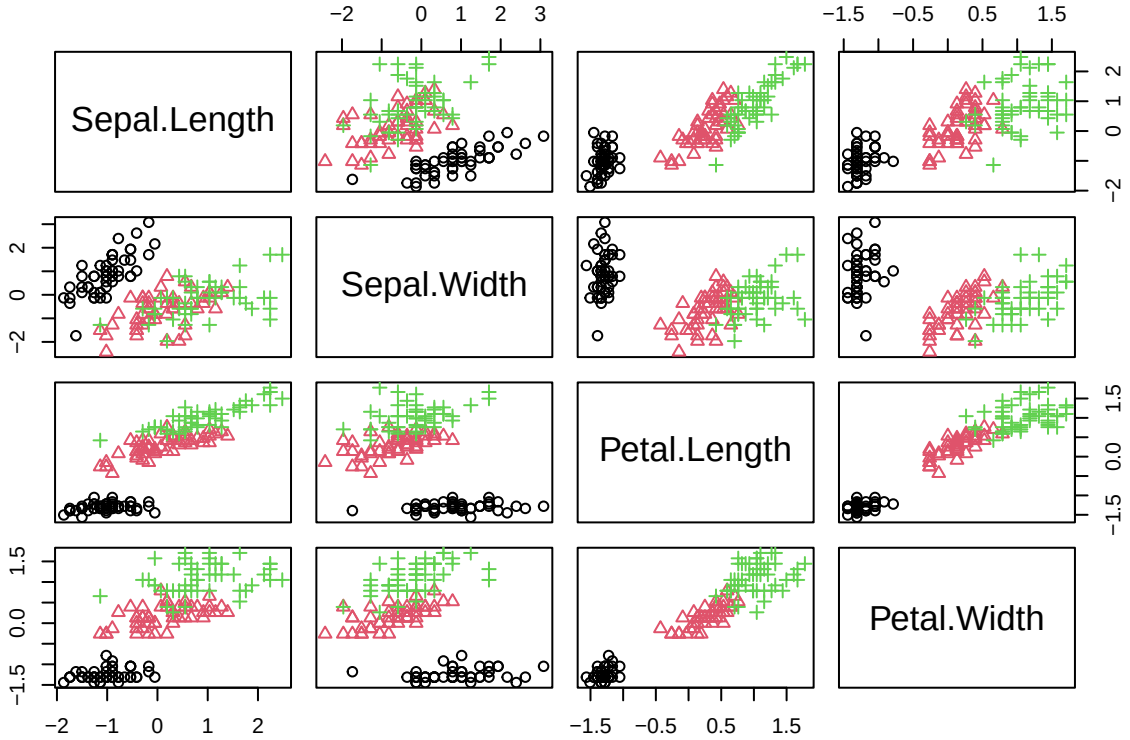
```
names(iris)
```

```
## [1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
```

```
Xmatrix<-scale((iris[,1:4]),scale=TRUE)
Ymatrix<-iris[,5]
table(Ymatrix)
```

```
## Ymatrix
##     setosa versicolor  virginica
##         50         50         50
```

```
pairs(Xmatrix,col=as.integer(Ymatrix),pch=as.integer(Ymatrix))
```




```
temp<-kmeans(Xmatrix,centers=3)
names(temp)
```

```
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
temp$centers #mean value of each cluster
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width
## 1  -0.05005221 -0.88042696    0.3465767   0.2805873
## 2   1.13217737  0.08812645    0.9928284   1.0141287
## 3  -1.01119138  0.85041372   -1.3006301  -1.2507035
```

```
temp$size #number of observations in each cluster
```

```
## [1] 53 47 50
```

```
temp$totss #SSE across all observations
```

```
## [1] 596
```

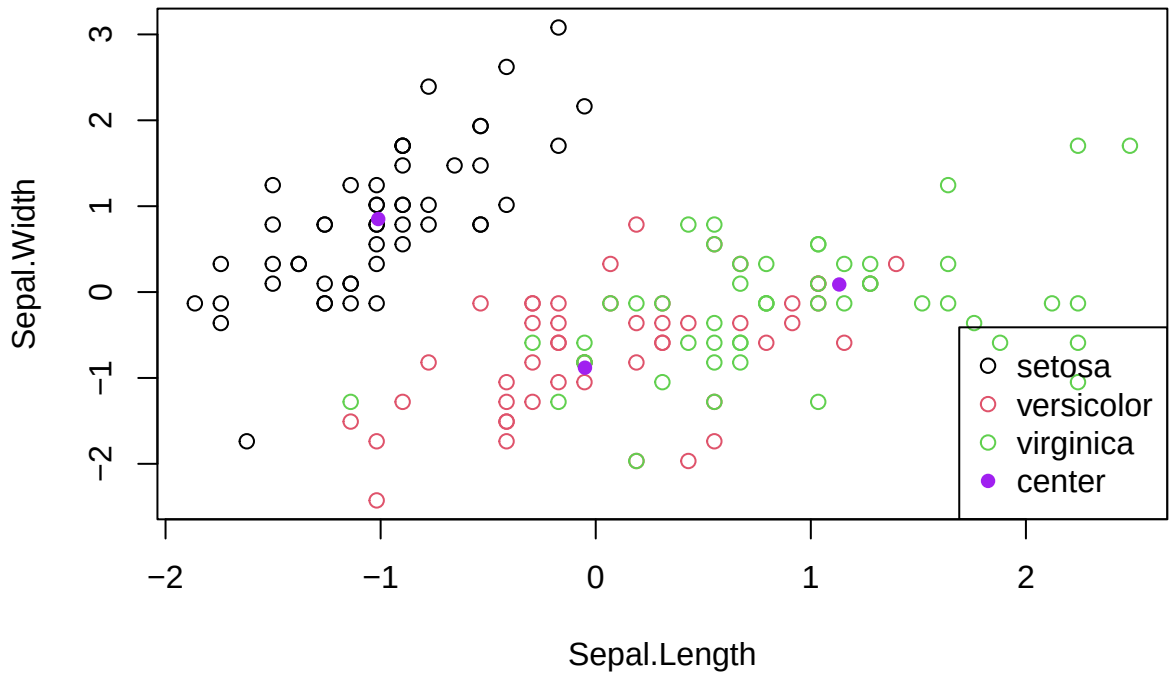
```
temp$tot.withinss #SSE across all observations in each cluster
```

```
## [1] 138.8884
```

```
temp$iter #Number of iterations used
```

```
## [1] 2
```

```
center<-temp$centers
plot(Xmatrix,col=as.integer(Ymatrix),pch=1)
points(center,pch=16,col="purple")
legend("bottomright",legend=c(levels(Ymatrix),"center"),col=c(1,2,3,"purple"),pch=c(1,1,1,16))
```




```
center
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width
## 1  -0.05005221 -0.88042696    0.3465767   0.2805873
## 2   1.13217737  0.08812645    0.9928284   1.0141287
## 3  -1.01119138  0.85041372   -1.3006301  -1.2507035
```

```
temp2<-as.matrix(temp$centers)
temp3<-t(temp2)
temp4<-matrix(0,nrow=nrow(temp3),ncol=ncol(temp3))
temp5<-temp4
```

```
x<-order(temp3[1,])
for (i in 1:3){
    temp4[2,x[i]]<-temp3[2,i]
    temp4[1,x[i]]<-temp3[1,i]
}
print(temp4)
```

```
##            [,1]       [,2]        [,3]
## [1,] 1.13217737 -1.0111914 -0.05005221
## [2,] 0.08812645  0.8504137 -0.88042696
## [3,] 0.00000000  0.0000000  0.00000000
## [4,] 0.00000000  0.0000000  0.00000000
```

```
for(i in 1:3){
    temp5[,i]<-colMeans(Xmatrix[Ymatrix==levels(Ymatrix)[i],])
}
temp5
```

```
##            [,1]       [,2]       [,3]
## [1,] -1.0111914  0.1119073  0.8992841
## [2,]  0.8504137 -0.6592236 -0.1911901
## [3,] -1.3006301  0.2843712  1.0162589
## [4,] -1.2507035  0.1661774  1.0845261
```

```
colnames(temp5)<-levels(Ymatrix)
temp5-temp4
```

```
##          setosa versicolor virginica
## [1,] -2.1433688  1.1230987 0.9493363
## [2,]  0.7622873 -1.5096373 0.6892368
## [3,] -1.3006301  0.2843712 1.0162589
## [4,] -1.2507035  0.1661774 1.0845261
```

```
table(Ymatrix,temp$cluster)
```

```
##
## Ymatrix       1  2  3
##   setosa      0  0 50
##   versicolor 39 11  0
##   virginica  14 36  0
```

```
names(temp)
```

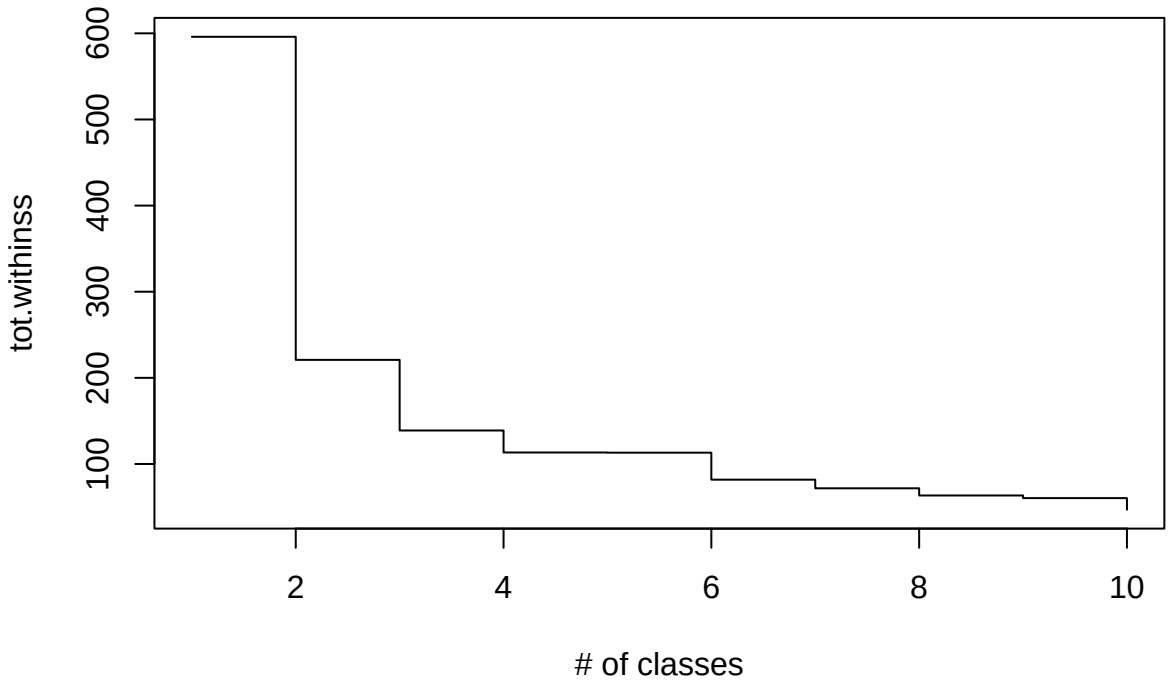
```
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
temp6<-matrix(0,nrow=10,ncol=2)
for (i in 1:10){
    temp<-kmeans(Xmatrix,centers=i)
    temp6[i,1]<-temp$tot.withinss
    temp6[i,2]<-temp$betweenss
}
colnames(temp6)<-c("tot.withinss","betweenss")
print(temp6)
```

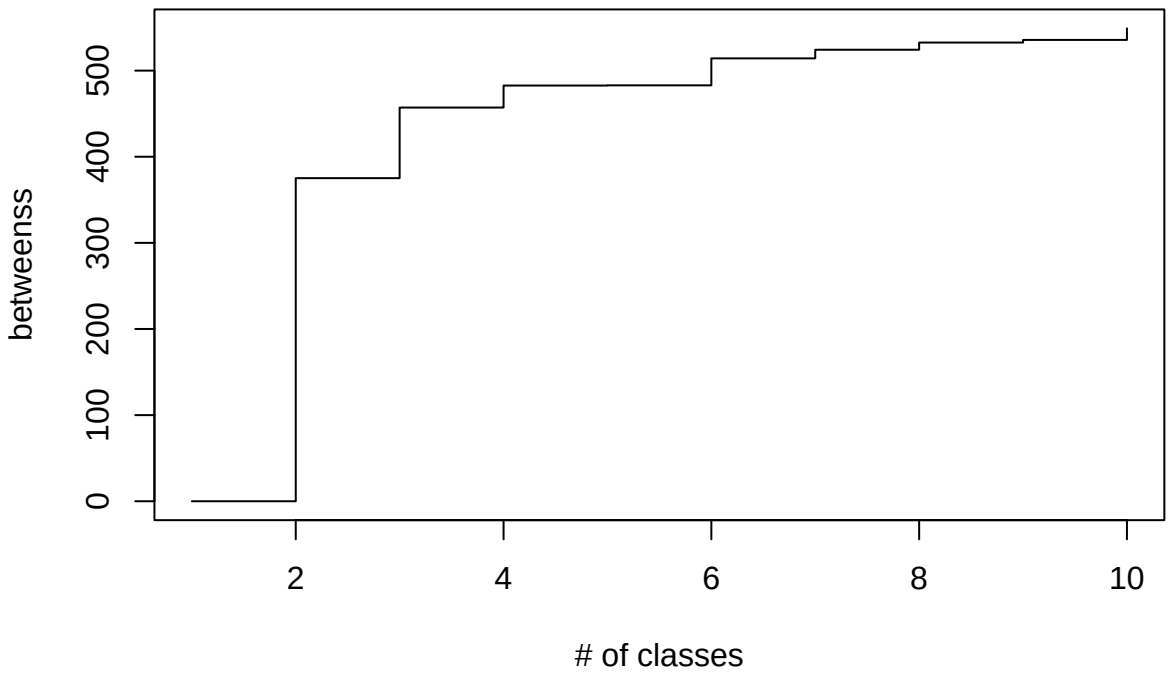
```
##       tot.withinss    betweenss
##  [1,]    596.00000 1.136868e-12
```

```
##  [2,]    220.87929 3.751207e+02
##  [3,]    138.88836 4.571116e+02
##  [4,]    113.33162 4.826684e+02
##  [5,]    113.12553 4.828745e+02
##  [6,]     81.76969 5.142303e+02
##  [7,]     71.75189 5.242481e+02
##  [8,]     63.43104 5.325690e+02
##  [9,]     60.34785 5.356521e+02
## [10,]     46.92474 5.490753e+02
```

```
plot(1:10,temp6[,1],xlab="# of classes",ylab="tot.withinss",type="s")
```




```
plot(1:10,temp6[,2],xlab="# of classes",ylab="betweenss",type="s")
```



The results from this problem match the conclusion found in Problem 6.2.